UNIVERSITY OF CAPE TOWN

Financial Analysis and Portfolio Management
Final Examination - 27 May 2019
Portfolio Optimization: ECO 4122Z

3 Hours

Maximum Marks: 100

INSTRUCTIONS TO CANDIDATES

i. Answer ALL questions

ii. Show all calculations

iii. Write legibly using blue or black ink

Additional Material: Formulae Sheet

1. Define the following terms:
   (a) first order stochastic dominance [2 marks]
   (b) second order stochastic dominance [2 marks]

   Consider two assets which will deliver a one-year return r_i on asset i with probabilities as set out below:

Asset	$P(r_i = -10\%)$	$P(r_i = -6\%)$	$P(r_i = 0\%)$	$P(r_i = +6\%)$	$P(r_i = +10\%)$
Asset 1	0.1	0.3	0.2	0.3	0.1
Asset 2	0.3	0.2	0.1	0.2	0.2

   (c) Compare the two assets in terms of first-order stochastic dominance [4 marks]

   [Total: 8 marks]

2. (a) In the context of mean-variance portfolio theory, define the term 'efficient frontier'. [2 marks]

   (b) Suppose that there are two risky assets: A and B, with the following properties.

	Expected Return	Standard Deviation
Asset A	0.08	0.2
Asset B	0.13	0.4

   The risk-free rate is 5 percent, and assets A and B have correlation of 0.625. Determine the expected return and standard deviation of the portfolio of A and B that maximizes the Sharpe ratio (i.e. the optimal portfolio). [10 marks]

   [Total 12 marks]

3. Suppose that Linda invests R60,000 in a stock with a beta of 1.2 and R40,000 in a stock with a beta of 0.9. If the riskfree rate is 6%, the market expected return is 12% and the CAPM holds, what must the expected return on Linda's portfolio be? [4 marks]

4. Consider the following information about two stocks (X and Y) and two common risk factors (1 and 2):

Stock	b_{i1}	b_{i2}	$E(R_i)$
X	1.25	3.2	14.25%
Y	2.4	2.5	16.6%

   (a) Assuming that the risk-free rate is 6.5%, calculate the levels of the factor risk premia that are consistent with the reported values for the factor betas and the expected returns for the two stocks. [6 marks]

   (b) You expect that in one year the stock prices for X and Y will be R600 and R450, respectively, none of the companies is expected to pay a dividend over the next year. Determine the expected current price of each company. [4 marks]

[Total 10 marks]

5. (a) Describe the primary purpose of an investment policy statement (IPS). [3 marks]

   (b) List any four typical restrictions that might be included within an IPS for a global equity portfolio for a charity fund. [4 marks]

[Total 7 marks]

6. The trustees of a pension fund are conducting their regular review of a fund manager's performance. The fund is invested in domestic equities. Amongst other things they have been supplied with the following data:

Review period (years)	1	3	5
Fund return p.a. (net of fees)	-11.01%	2.06%	5.16%
Index return p.a.	-9.73%	2.30%	5.06%
Tracking Error	2.16%	1.15%	0.88%

   (a) Describe the key features of these results. [2 marks]

   (b) State what other information you would require in order to make a decision regarding whether the fund manager should be retained. [6 marks]

   A newly appointed trustee has proposed switching the funds to a fund manager who specialises in newly emerging stock markets as she has heard that these markets have shown strong growth recently.

   (c) Comment on this proposal. [3 marks]

   (d) Suggest any problems that might be encountered if it were adopted. [3 marks]

[Total 14 marks]

7. Consider the following trading and performance data for two different equity mutual funds:

	Fund 1	Fund 2
Assets under Management, Avg. for Past 12 months (mil)	ZAR289.4	ZAR5,567.3
Security Sales, Past 12 months (mil)	ZAR37.2	ZAR437.1
Expense Ratio	0.33%	0.21%
Pretax Return, 3-year avg.	9.98%	9.83%
Tax-adjusted Return, 3-year avg.	9.43%	9.54%

   (a) Calculate the portfolio turnover ratio for each fund. [4 marks]

   (b) Calculate the tax cost ratio for each fund. [6 marks]

   (c) Which funds were the most and least tax efficient in the operations? Why? [5 marks]

[Total 15 marks]

8. Two investment managers each managing part of a large pension scheme have produced the following figures:

	Fund (ZAR million	Equities	Bonds	Real Estate	Cash
A	100	25	50	20	5
B	100	35	40	15	10

Constituents of Index (%)

	Equities	Bonds	Real Estate	Cash
Index	30%	50%	20%	0%

Performance in year	Equities	Bonds	Real Estate	Cash
A	6.0%	3.0%	7.0%	0.5%
B	4.5%	4.0%	6.5%	0.5%
Index	5%	2%	6%	

(a) Conduct an attribution analysis for the two funds in year 1. [6 marks]

The following attribution analysis has been produced for years 2 and 3.

Year 2	Fund Performance	Index Performance	Sector Selection	Stock Selection
A	8.0%	7.0%	-0.5%	1.5%
B	6.0%	7.0%	0.0%	-1.0%

Year 3	Fund Performance	Index Performance	Sector Selection	Stock Selection
A	1.5%	3.0%	0.5%	-2.0%
B	4.5%	3.0%	2.0%	-0.5%

Both managers are operating under the same mandate with the objective of outperforming their benchmark by 1% p.a. over a three year period.

(b) Comment on the performance over the three year period. [4 marks]

Currently the benchmark is rebalanced every three years and the fund managers can deviate from the benchmark as they see fit. The trustees have now decided to rebalance the benchmark quarterly and to limit the freedom of the fund managers to deviate from the benchmark.

(c) Discuss the advantages and disadvantages of these proposed changes. [5 marks]

[Total 15 marks]

9. The following information relates to the performance of two investment trusts and their equivalent benchmark index over a three year period. The annual risk-free rate of return over this period was 4% per annum.

Question 9 continues on the next page. . .

	Trust A	Trust B	Index
Annual return (% p.a.)	9.0	8.0	7.0
Standard deviation (% p.a.)	13.5	9.5	6.5
Correlation coefficient with index	0.36	0.75	1.00

(a) Calculate the Treynor measure, Sharpe ratio, Jensen's alpha and M^2 composite performance measures for each trust. [10 marks]

(b) Comment on the results from part (a), stating any limitations that apply to them. [5 marks]

[Total 15 marks]